



Maximum sub-array-sum

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Given an array of n integers $x_1, x_2, \dots, x_n$, our task is to find the maximum sub array sum, i.e., For example, in the array $[-1, 2, 4, -3, 5, 2, -5, 2]$ the following subarray produces the maximum sum 10: $[-1, 2, 4, -3, 5, 2, -5, 2]$

Input Format

Two line input First line should read an integer value which represents size of array Second line consists list of elements(integers) separated by space

Constraints

1

Output Format

It prints integer value which represents maximum sub array sum

Sample Input 0

```
8
-1 2 4 -3 5 2 -5 2
```

Sample Output 0

```
10
```

Sample Input 1

```
7
-3 4 -1 2 1 -4 3
```

Sample Output 1

```
6
```

Sample Input 2

```
8
-1 2 4 -3 5 2 -5 2
```

Sample Output 2

```
10
```

Sample Input 3

```
6
-1 2 3 4 5 -15
```

Sample Output 3

14

Sample Input 4

7
-3 4 -1 2 1 -4 3

Sample Output 4

6

Sample Input 5

5
1 2 3 4 -10

Sample Output 5

10

Change Theme

Language

C



```
2  #include <string.h>
3  #include <math.h>
4  #include <stdlib.h>
5
6  int main() {
7      int n;
8      scanf("%d", &n);
9      int nums[n];
10     for(int i=0;i<n;i++){
11         scanf("%d", &nums[i]);
12     }
13     int currentsum = nums[0];
14     int maxsum = nums[0];
15     for(int i=1;i<n;i++){
16         if(currentsum + nums[i] > nums[i]){
17             currentsum += nums[i];
18         }
19         else{
20             currentsum = nums[i];
21         }
22         if(currentsum > maxsum){
23             maxsum = currentsum;
24         }
25     }
26     printf("%d", maxsum);
27     return 0;
28 }
29
```

Line: 29 Col: 1

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Test against custom input

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